

ESP32 LoRa GATEWAY For Data Monitoring

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In a boiler safety project, we collect 32 thermocouple readings from the depth of a 500MW boiler second pass area, where ambient temperature on a normal day is 65 to 70°C due to hot area leakages in the vicinity. However, the thermocouples need to measure temperatures ranging from 350 to 150°C, depending on their positions.

During a boiler shutdown, we fixed the thermocouples in their strategic places, connected an Arduino Mega board, fixed the LoRa radio with only its antenna sticking outside the IP64 cabinet, and then fixed the box on a post. The 220V AC supply from a nearby pillar was converted through a smartphone adaptor to 5V DC and introduced into the cabinet through a small hole below it.

The two sets of 16 each K-type thermocouples having 32 wires were introduced into the two boxes through side slots, which were then sealed with special insulating putty and m-seal. The thought of providing an LED indication to show health of these devices was turned down as hardly anyone visits the place at this height and high temperature. We finally have two black boxes fixed on

a pillar with 32 thermocouple cables entering each box.

Who uses this data?

All the data goes every ten seconds to a nearby service building situated a good 700 metres away where the boiler maintenance team has their office. The small handheld Arduino Mega device, besides storing the data in CSV format, shows it on a small TFT screen in four columns. It also has capabilities to demonstrate

the rate-of-rise of temperature with audio/visual alarm. The data arrives in a CSV format, as shown in Fig. 2.

The data was being monitored by the boiler maintenance department as they are the sole agency to take care of any abnormalities of the parameters. But one day an important aberration occurred when the maintenance staff was on holiday. So, it was decided that the data should be posted on LAN/WAN so that it could be monitored

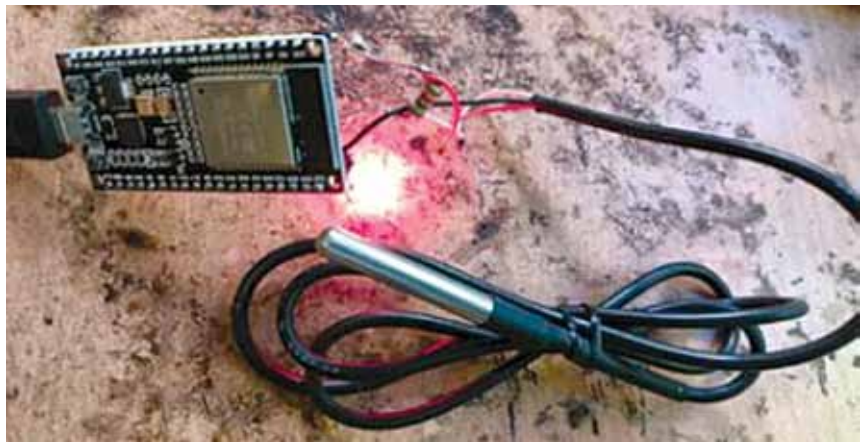


Fig. 1: Author's prototype

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Fig. 2: The data in CSV format

Bill of Material

1. ESP32 module -1
2. LoRa module -1
3. LoRa antenna -1
4. Breadboard
5. Data transfer cable for ESP32
6. Jumper wires
7. Power supply